Online Supplement: Access and Selection in Public and Private Innovation Intermediation

Equation and table references without an S-prefix refer to the main manuscript. The proofs retain the original primitives and benchmark certification scope.

S.1 Proof details and boundary checks

S.1.1 Research investment and Cournot continuation

Taking the successful-new-good Cournot quantities as fixed after successful commercialisation, a matched founder pair solves $\max_{e\geq 0}(p_0+\zeta e+\lambda\theta_A\theta_B)\Pi-ke^2/2$. The first-order condition $\zeta\Pi-ke=0$ yields $e^*=\zeta\Pi/k$, with strict second derivative $-k<0$. Substitution gives $R_0+\lambda\Pi\theta_A\theta_B$ as defined in the main text. This calculation assumes the admissible interior success-probability domain, not a corner where $p=1$. Incumbent Cournot quantities follow by solving the simultaneous linear first-order conditions (2); all output and positive-definiteness conditions must hold.

S.1.2 Global private-only posted fee, including kink

Holding (m_j,q_j) fixed, private-only total fee receipts divided by N_i equal f for $0\leq f\leq U_i$, $f(U_i+T_i-f)/T_i$ for $U_i\leq f\leq U_i+T_i$, and zero for $f\geq U_i+T_i$. The first piece is increasing; the second is strictly concave with unconstrained maximum $(U_i+T_i)/2$. This point belongs to the sorting interval exactly when $U_i\leq T_i$. Otherwise the global maximiser is the full-participation corner $f_i^*=U_i$. The function is continuous at the kink. A positive basic benefit $U_i>0$ rules out zero-revenue opt-out as the optimum.

For co-provision, only the premium difference matters: for $f\in[0,\Delta_i]$, $\mathcal{R}_i(f)/N_i=f(1-f/\Delta_i)$ is strictly concave with maximum $f=\Delta_i/2$. It yields premium share $1/2$ and direct real benefit $\mathbb{E}[u_i+\Delta_i\theta_i\mathbf{1}\{\theta_i\geq 1/2\}]=u_i+3\Delta_i/8$. Public basic participation remains full regardless of premium demand.

S.1.3 Global private/private fee continuation at the three baseline calibrations

Unlike the single-private-provider case, a fee change under (P,P) changes the other jurisdiction's participation cutoff. The following argument establishes that *the three numerical examples used in Table 1* have an admissible, unique global pure Nash equilibrium in nonnegative posted fees. The required coefficient inequalities are verified by exact rational arithmetic for each example; they are not assumed for arbitrary parameters.

Write $x_i=\ell_i$ for the local cutoff, $a_i=a_i^m$, $b_i=b_i^m$, and use the frozen baseline $\Delta_i=1$ with $u_i=u_i(P)>0$. Define

$$D_i(y)=1+\tfrac{b_i}{2}(1-y^2),\qquad h_i=a_i+b_i,\qquad z=h_Ah_B.$$

The simultaneous founder participation mapping at any nonnegative fee pair is

$$x_i = \left[\frac{f_i - u_i - a_i(1 - x_j)}{D_i(x_j)}\right]_0^1. \tag{S.1}$$

On an interior branch, the derivative with respect to x_j is $(a_i + b_i x_i x_j)/D_i(x_j) \le h_i$. On clipped branches it is zero. Each of the three frozen parameter sets satisfies $\max_i h_i < 1$. Consequently the coupled map is a contraction on $[0,1]^2$, so the founder-cutoff fixed point exists and is unique *for every nonnegative fee pair*, including out-of-equilibrium fees.

Fix f_j and use $x = x_i$ as the local decision variable. The opponent cutoff conditional on this cutoff is uniquely

$$y(x) = \left[\frac{f_j - u_j - a_j(1 - x)}{D_j(x)}\right]_0^1.$$

Inverting the local participation equation gives an implementable posted fee

$$f_i(x; f_j) = u_i + a_i(1 - y(x)) + D_i(y(x))x.$$

Where the opposing cutoff is interior, $y'(x) = (a_j + b_j xy)/D_j(x)$; at flat clipped branches the derivative is zero. Both one-sided derivatives at any kink satisfy

$$1 - z \;\le\; f_i'(x) \;\le\; 1 + b_i/2. \tag{S.2}$$

The inverse fee is strictly increasing since $z < 1$. Fees inducing full participation are weakly dominated by the maximal fee retaining it; fees inducing zero participants yield no revenue and cannot be optimal. Thus all global fee best replies are obtained by maximising $N_i f_i(x; f_j)(1-x)$ over $0 \le x \le 1$. Its derivative signs at the outer cutoff ranges and the bounds $u_i + x \le f_i(x; f_j) \le u_i + a_i + (1 + b_i/2)x$ give the *global* best-response cutoff bounds

$$L_i = \frac{1 - z - u_i - a_i}{2 + b_i/2 - z}, \qquad U_i = \frac{1 + b_i/2 - u_i}{2 + b_i/2}, \qquad 0 < L_i < U_i < 1. \tag{S.3}$$

Every global fee best reply lies in

$$K_i = [\, u_i + L_i,\; u_i + a_i + (1 + b_i/2)U_i \,].$$

Exact rational checks for all three calibrations give $u_i + a_i < \inf K_i \le \sup K_i < u_i + 1$. Hence for every opponent fee in K_j, its cutoff $y(x)$ is strictly interior throughout the *entire* own-cutoff deviation interval $x \in [0,1]$; neither full-coverage nor exit kinks are being suppressed in the global best-response test.

On that interval, $y''(x) = b_j[y + 2xy']/D_j(x) \ge 0$. Writing $c_i(x,y) = a_i + b_i xy$, we have

$$f_i'(x) = D_i(y) - c_i(x,y)y'(x) > 0, \qquad f_i''(x) = -b_i(2y + xy')y' - c_i(x,y)y'' \le 0.$$

Consequently

$$\frac{d^2}{dx^2}\{N_i f_i(x; f_j)(1 - x)\} = N_i\{(1 - x)f_i''(x) - 2f_i'(x)\} < 0,$$

so the global best reply is unique for every opponent fee in K_j. Differentiating the best-response first-order condition and using the preceding inequalities yields a Lipschitz bound for the fee best-response map:

$$B_i := h_i + \frac{(1 + b_i/2)\{b_i + b_i h_j + b_j h_i + h_i\}}{2(1 - z)} \quad \text{with} \quad B_i < 0.01 \tag{S.4}$$

in each of the three exact baseline calibrations. The map sends $K_A \times K_B$ continuously into itself and is a contraction; Banach's theorem implies a unique pure price equilibrium on that

rectangle. Because (S.3) captures *every* global best reply against any nonnegative rival fee, no additional pure-fee equilibrium exists outside the rectangle.

For an independently checkable exact-rational enclosure, define

$$C_i(x_A, x_B) = a_i + b_i x_A x_B, \qquad f_i = u_i + a_i(1 - x_j) + D_i(x_j)x_i,$$

and the two interior stationary equations

$$\Phi_i(x_A, x_B) = (1 - x_i)\left(D_i(x_j) - \frac{C_i C_j}{D_j(x_i)}\right) - f_i = 0.$$

For each calibration, the source `stage11r_pp_global_certificate.py` evaluates all four rational interval faces of a cutoff rectangle of radius 10^{-9} around the recorded decimal-rational candidate; the lower and upper own-coordinate faces have strict positive and negative signs, respectively. This locates an exact equilibrium root by the Poincaré–Miranda theorem. The global argument above establishes that it is the unique P/P pure price Nash in the whole nonnegative fee strategy space.

Finally the script reconstructs *all* sixteen first-stage entry configurations, all four independent government/provider flips at each profile, and all fifteen alternatives to the restricted planner's selected profile, using interval enclosures for the P/P fees and exact rational arithmetic elsewhere. All selected baseline NE inequalities, all non-NE profitable deviation witnesses, and all planner welfare gaps have strict signs. In the $\lambda = 0.5$ skill case the smallest selected entry slack is greater than 0.0000316701024488 at the printed precision. The reproducible source and its complete statement of proof scope are archived in the Stage 11R-P0 verification report. None of the results supplies a general theorem for other skill distributions, parameters or competition forms, and none is encoded as a full-game Lean proof.

S.1.4 Separate participation mass and skill moment

Uniform ability gives $m_i = \int_{\ell_i}^1 d\theta = 1 - \ell_i$, $q_i = \int_{\ell_i}^1 \theta\, d\theta = (1 - \ell_i^2)/2$. A random draw of the same mass has expected first moment $m_i/2$; subtracting yields $\ell_i(1-\ell_i)/2$. This is a same-mass sorting comparison, not a statement that equilibrium quality-weighted matches always increase when private fees rise.

S.1.5 The conditional public-addition wedge

With the other region fully participating, moving region i from P to B changes its direct real benefit by $N_i[u_i + 3\Delta_i/8 - (u_i(1 - \ell_i) + \Delta_i(1 - \ell_i^2)/2)]$, which is H_i in (28). The matching count and ability products change by $C\ell_i$ and $C\ell_i^2/4$. Multiplying these by S_i^0 and λK_i gives the local threshold. Repeating for region j gives the foreign increment (30). No planner choice over tariffs, research effort or private contracts is implied.

S.1.6 Binary-entry verification protocol

For any local status s_i, G_i's unilateral deviation toggles $0 \leftrightarrow G$ or $P \leftrightarrow B$; P_i's unilateral deviation toggles $0 \leftrightarrow P$ or $G \leftrightarrow B$. Each profile passes the first-stage pure Nash check if and only if all four relevant payoff improvements are nonpositive *for the selected admissible downstream continuations*. When any continuation is set-valued, the selection rule must be stated. A finite numerical grid of private fees does not prove unrestricted global fee equilibrium.

S.1.7 The binding public-cost thresholds and exact open-region check

For the frozen $(s_A, s_B) = (B, P)$ and (B, B) profiles, changing only region B's public fixed cost does not affect any fee or founder participation continuation. Conditional on the same

technology scenario s, define the public-entry threshold by the exact reduced-welfare difference computed at zero F_{G_B}:

$$T_s = W_B((B,B); F_{G_B} = 0, s) - W_B((B,P); F_{G_B} = 0, s). \tag{S.5}$$

The local public-addition margin at any F_{G_B} is exactly $T_s - F_{G_B}$. In the maintained full-coverage opposite region, Proposition 3 equivalently gives $T_s = H_B + C(S_B^0 \ell_B + S_B^1 \ell_B^2/4)$, where the fee-selected ℓ_B is the appropriate private-only continuation for that technology scenario. The homogeneous comparison uses its own recentered matching coefficients; it is not obtained by reusing the heterogeneous cutoff.

In exact arithmetic the ordering is $T_{\text{pooled},\,\lambda=0.5} < T_{\text{skill},\,\lambda=0} < T_{\text{skill},\,\lambda=0.5}$. The simultaneous three-case pure Nash pattern thus requires $T_{\text{skill},0} < F_{G_B} < T_{\text{skill},0.5}$. It is *sufficient* for the three particular calibrations because the independent rational checker enumerates all sixteen provision states, four government/private binary flips in each state, and fifteen planner competitors, verifying strict signs on eight adjoining subintervals of the full threshold-defined closed cost band. Only the two known G_B comparisons tie at the exterior endpoints. Their exact affine $T_s - F_{G_B}$ expressions make the remaining inequalities strict in the open interval, with no appeal to finite-grid sampling.

The same verifier recomputes the global P/P sufficient conditions and all sixteen states using rational inclusion arithmetic when F_{G_A}, F_{G_B} and ω_B vary jointly within the explicit rectangle in (37), subject to $\omega_A = 1 - \omega_B$. The exact four-face fee-root checks include the entire parameter box and all provider-entry conditions remain strictly signed. This shows that the example persists on a small multidimensional open set, while no distribution-wide or different-conduct robustness theorem is implied.

S.1.8 Private fee receipts and why co-provision need not benefit private owners

Let $U_i > 0$ and $T_i = \Delta_i + b_i^m q_j \geq \Delta_i$, holding the other-region participant moments fixed. In the interior monopoly service regime, private-only receipts equal $N_i(U_i+T_i)^2/(4T_i)$; under free-basic co-provision, receipts equal $N_i\Delta_i/4$. Their difference is

$$\mathcal{R}_i^P - \mathcal{R}_i^B = \frac{N_i}{4}\left(\frac{(U_i+T_i)^2}{T_i} - \Delta_i\right) > 0 \tag{S.6}$$

since $(U_i+T_i)^2/T_i > T_i \geq \Delta_i$. In the full-coverage branch, receipts are $N_i U_i$ with $U_i \geq T_i \geq \Delta_i$, which also strictly exceeds $N_i\Delta_i/4$. The fixed private cost cancels. The result compares a *given active private provider* under the two arrangements, and says nothing about whether local public entry is welfare-improving. It also does not imply a private supplier exits: in the co-provision continuation the private entrant earns $N_i\Delta_i/4 - F_{P_i}$, which may be strictly positive. This is an additional *expositional implication* of the already frozen fee formulas and is not advertised as a new general theorem.

S.1.9 Fixed participation: how technical selection changes project returns

For the uniform distribution and a common participant cutoff ℓ_i, the mass is $m_i = 1-\ell_i$ and skill moment $q_i = (1-\ell_i^2)/2$. Suppose region j is fully covered, so $(m_j, q_j) = (1, 1/2)$, and compare the heterogeneous and mean-homogenised innovation premiums at the *same* participation decision. The full-coverage normalisation of the homogeneous increment implies its matching effect is $C\lambda K_i m_i/4$. The heterogeneous effect is $C\lambda K_i q_i/2$. Their difference is

$$\frac{C\lambda K_i}{4}(2q_i - m_i) = \frac{C\lambda K_i}{4}\left[(1-\ell_i^2) - (1-\ell_i)\right] = \frac{C\lambda K_i}{4}\ell_i(1-\ell_i). \tag{S.7}$$

Therefore the sign is not necessarily beneficial for either region; it equals $\operatorname{sign}(K_i)$ for $\lambda > 0$ and $0 < \ell_i < 1$. The comparison should not be silently imported into the reoptimised P-only continuation, where the two distributions imply different optimum fees and cutoffs.

S.1.10 Reduced participation-externality representation

For every fixed valid selected continuation, put $\mathcal{A}_i = C(p_b M_i + \omega_i R_0)$ and $\mathcal{B}_i = C\lambda(M_i + \omega_i \Pi)$, so

$$W_i(s) = W_i^0 + Z_i(s) - F_{G_i} g_i - F_{P_i} p_i + \mathcal{A}_i m_A(s) m_B(s) + \mathcal{B}_i q_A(s) q_B(s). \tag{S.8}$$

This recentering makes the relation to ordinary multi-region public-good externality games explicit, rather than asserting a new welfare-wedge theorem. The endogenous map from strategic fees and entry into (m_i, q_i) is the model-specific object. It does not by itself disprove potential nesting in a more general predecessor's theorem.

S.2 Reproducibility and certification boundaries

The source and exact verification scripts are supplied in the accompanying replication materials. Identifying repository links are omitted from the anonymous review version. The frozen theory is identified by `STAGE33-2026-10-28-THEORY-FREEZE-v1`; its recorded baseline is reproduced with the Python-standard-library script `stage33_reproduce.py` (in the `production/verification` directory of the `hotelling_entry_experimentation` research tree). The original implementation runs sixteen entry configurations for each of three reported technology scenarios, checks all four first-stage unilateral binary flips, and computes the restricted planner under its numerical P/P fee continuation. A separate standard-library verifier in the Stage 11R-P0 review package now independently reconstructs the baseline parameter values, checks the price-box and contraction inequalities using exact fractions, provides rational root enclosures and evaluates all sixty-four first-stage flips in each scenario. The former fee-grid check is no longer the sole evidence for the three reported P/P continuations. The analytic conditions are parameter-specific and the associated machine checks are not a proof-assistant formalisation.

The separately maintained Lean 4 certificate checks eighteen theorem statements covering selected proof-critical inequalities and identities under their specified hypotheses. It does *not* encode the entire Cournot–research–participation–entry game or the analytic global P/P fee lemma just supplied. The latter has an independent mathematical argument and exact rational baseline verification; its broader portability is not claimed. Algebraic proof, computer-assisted interval verification and Lean theorem verification are distinct evidentiary categories.

The Beta$(2, 2)$ and differentiated-Bertrand alternatives are negative portability records. They are not successful robustness tests. Untested policy instruments remain outside the analysis.

S.3 The repriced diagnostic comparison

Under complete opposite-region participation, heterogeneous founder willingness to pay has intercept U and slope $d + h$. Pooled success adds $h/2$ to the intercept and leaves slope d. Applying the global interior/full-coverage pricing formula gives (31). If both cutoffs are positive, subtracting gives (32). If only one is zero, it must be ℓ_L: $U \geq d+h$ would also imply $U+h/2 \geq d$. Thus $\ell_H \geq \ell_L$ everywhere, strictly for $h > 0$ unless both are zero.

For the heterogeneous technology Proposition 3 gives

$$T_H = N_i[u_i\ell_H - d/8 + d\ell_H^2/2] + CS_i^0\ell_H + C\lambda K_i\ell_H^2/4.$$

For pooled success the matching increment is constant per pair, so the corresponding expression is

$$T_L = N_i[u_i\ell_L - d/8 + d\ell_L^2/2] + CS_i^0\ell_L + C\lambda K_i\ell_L/4.$$

Subtract, factor $\ell_H^2 - \ell_L^2 = (\ell_H - \ell_L)(\ell_H + \ell_L)$, and obtain (33). The formula remains valid when a cutoff is zero: adding a public alternative may then forgo some direct premium value, reflected in the same $-N_i d/8$ term. No division by a possibly signed local market coefficient is used.

On an interior branch, $h = \eta_i\lambda$ and $\ell_H = 1/2 - U/[2(d + h)]$, with U constant in λ. Differentiating gives $\ell_H' = U\eta_i/[2(d + h)^2]$ and then (34). These are identities conditional on the opposite region's full participation, not unrestricted four-entry equilibrium comparative statics. The additional exact checker reconstructs the three original thresholds independently of numerical fee searches, verifies the decomposition, and reports the direct-service and market-incidence differences without changing any primitive.

For the analytical negative-sign case, $\omega_i = 0$ implies $a_i^m = b_i^m = 0$, so $U = u_i$, $h = 0$, $\ell_H = \ell_L = \ell$ and $K_i = M_i$. Substitution in (33) gives $C\lambda M_i\ell(\ell - 1)/4 < 0$. The cutoffs and threshold difference are continuous in ω_i on this strict interior price domain; negativity therefore persists for an interval of positive profit shares. If also $0 < u_i < 2d/3$, the common direct-service threshold term is $N_i u_i(2-3u_i/d)/8 > 0$. Since the local matching terms are positive for $M_i > 0$, both public thresholds are then positive: the opposite preference can concern a positive public cost. This is a conditional parameter region within the maintained model, not a new numerical calibration or a full first-stage equilibrium theorem.